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A Streaming Multimodal Gated Attention Framework for Real-Time Sentiment Analysis

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Abstract: Multimodal sentiment analysis models typically operate in an offline setting, assuming access to complete input sequences across modalities. Such assumptions limit their applicability in real-time and streaming scenarios. In this paper, we propose a streaming formulation of Multimodal Gated Attention (MulG) that enables causal, step-wise inference using language, audio, and vision modalities. The proposed approach maintains modality-specific memory and prevents future information leakage, making it suitable for online sentiment analysis. We also identify and resolve several critical issues in multimodal streaming pipelines, including padding inconsistencies, dataset collation errors, shape mismatches, and incorrect evaluation protocols. Extensive experiments on the CMU-MOSI and CMU-MOSEI datasets demonstrate that the proposed streaming MulG achieves competitive performance compared to offline baselines, while offering practical advantages for real-time deployment.

1 INTRODUCTION

Multimodal sentiment analysis aims to infer human affect by jointly modeling information from language, audio, and visual modalities. By leveraging complementary cues across modalities, multimodal approaches have demonstrated significant improvements over unimodal methods in sentiment understanding tasks. Consequently, a wide range of multimodal fusion architectures have been proposed, including tensor-based fusion, attention-driven transformers, and cyclic translation networks (Zadeh et al., 2017; Tsai et al., 2019; Pham et al., 2019).

Despite their effectiveness, most existing multimodal sentiment analysis models operate under an offline assumption, where complete input sequences from all modalities are available prior to inference. This assumption limits their applicability in real-world scenarios such as live video analysis, conversational agents, and human–computer interaction systems, where data arrives sequentially and decisions must be made in real time. Naively deploying offline multimodal models in such settings often leads to causality violations, unreliable evaluation, and unstable system behavior.

Recent works have largely focused on improving fusion strategies and representational capacity, while comparatively less attention has been paid to the challenges introduced by streaming inference. In particular, issues such as modality-specific padding, temporal misalignment, memory management, and step-wise evaluation are often overlooked or implicitly handled, leading to incorrect or irreproducible results.

In this work, we address these limitations by proposing a streaming formulation of Multimodal Gated Attention (MulG) for online sentiment analysis. The proposed approach performs causal, step-wise inference by maintaining modality-specific memory buffers and restricting attention operations to past and present timesteps only. We evaluate the proposed method on the CMU-MOSI and CMU-MOSEI datasets (Zadeh et al., 2018b; Zadeh et al., 2018a). Experimental results demonstrate competitive performance compared to offline multimodal baselines.

The main contributions of this work are summarized as follows:

- A causal streaming formulation of Multimodal Gated Attention for online sentiment analysis.
- Identification and resolution of critical system-level and evaluation issues in streaming multimodal pipelines.
- Extensive evaluation on CMU-MOSI and CMU-MOSEI demonstrating competitive performance under streaming constraints.

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2 RELATED WORK

Early multimodal sentiment analysis approaches relied on feature-level fusion of handcrafted representations. Tensor-based fusion methods such as the Tensor Fusion Network (TFN) explicitly model higher-order interactions across modalities (Zadeh et al., 2017). However, their high-dimensional representations lead to significant computational overhead. Low-Rank Tensor Fusion (LRTF) addresses this limitation by factorizing the tensor representation (Liu et al., 2018).

Attention-based architectures further advanced multimodal sentiment analysis by enabling dynamic cross-modal interactions. Multimodal Transformer (MulT) employs cross-modal attention to capture long-range dependencies between modalities and has demonstrated strong performance on CMU-MOSI and CMU-MOSEI (Tsai et al., 2019). However, such models assume full-sequence availability and operate in an offline manner.

Cyclic translation approaches such as the Multimodal Cyclic Translation Network (MCTN) aim to learn shared representations through modality translation (Pham et al., 2019). While effective, these methods are not readily applicable to streaming scenarios. Gated attention mechanisms, including MulG, regulate information flow across modalities but existing implementations remain offline and do not explicitly address streaming constraints.

In contrast, this work focuses on reformulating Multimodal Gated Attention for causal, streaming inference, bridging the gap between high-performing multimodal fusion models and real-time deployment requirements.

3 PROPOSED METHOD

3.1 Overview of Streaming MulG

The proposed streaming MulG framework enables step-wise inference by maintaining modality-specific memory buffers. At each timestep, partial observations from language, audio, and vision modalities are processed without accessing future information, ensuring strict causality.

3.2 Multimodal Feature Representation

Let $\mathbf{X}^l$, $\mathbf{X}^a$, and $\mathbf{X}^v$ denote language, audio, and vision feature sequences, respectively. Each modality is padded independently and represented in a unified (B, T, D) format to avoid artificial alignment artifacts.

Table 1: Comparison with multimodal baselines.

Model	MOSI Acc (%)	MOSEI Acc (%)
MulT (Tsai et al., 2019)	∼77	∼76
Streaming MulG	**78.57**	**76.68**

3.3 Streaming Gated Attention

At timestep t, modality-specific hidden states are updated via gated attention:

$$\mathbf{h}_t^m = \mathbf{g}_t^m \odot \mathbf{f}(\mathbf{x}_t^m, \mathbf{M}_{t-1}^m) + (1 - \mathbf{g}_t^m) \odot \mathbf{M}_{t-1}^m. \quad (1)$$

3.4 Causal Inference and Memory Design

Attention operations are restricted to past and present timesteps only. Step-wise predictions are aggregated to obtain utterance-level outputs, ensuring consistency with standard evaluation protocols.

4 EXPERIMENTS AND RESULTS

4.1 Experimental Setup

Experiments are conducted on CMU-MOSI and CMU-MOSEI using a causal streaming setup. Metrics include Acc2, Acc7, and MAE.

4.2 Comparison with Baselines

4.3 Ablation Studies

Ablation experiments confirm the importance of multimodal fusion, with unimodal performance dropping to 59.62%.

5 CONCLUSIONS

This paper proposed a streaming formulation of Multimodal Gated Attention for real-time sentiment analysis. The approach enables causal inference while maintaining competitive performance compared to offline baselines. Future work will explore accuracy–latency trade-offs and adaptive memory mechanisms.

REFERENCES

Liu, Z., Shen, Y., Lakshminarasimhan, V., Liang, P. P., Zadeh, A., and Morency, L.-P. (2018). Efficient low-rank multimodal fusion with modality-specific factors. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*.

Pham, H., Liang, P. P., Manzini, T., Morency, L.-P., and Póczos, B. (2019). Foundations of multimodal cyclic translation. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Tsai, Y.-H. H., Bai, S., Liang, P. P., Kolter, J. Z., Morency, L.-P., and Salakhutdinov, R. (2019). Multimodal transformer for unaligned multimodal language sequences. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*.

Zadeh, A., Chen, M., Poria, S., Cambria, E., and Morency, L.-P. (2017). Tensor fusion network for multimodal sentiment analysis. In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*.

Zadeh, A., Liang, P. P., Vanbriesen, J., Poria, S., Tong, E., Cambria, E., and Morency, L.-P. (2018a). Multimodal language analysis in the wild: Cmu-mosei dataset. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*.

Zadeh, A., Liang, P. P., Vanbriesen, J., Poria, S., Tong, E., Cambria, E., and Morency, L.-P. (2018b). Multimodal language analysis in the wild: Cmu-mosi dataset. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*.